# PRACTICE QUESTIONS DISCRETE RANDOM VARIABLES

## EXERCISE – 3.1

***DISCRETE PROBABILITY DISTRIBUTION***

**1.** Write out the probability distribution for the number of tails, X, obtained when three fair coins are tossed once.
$(Ans: see table)$

| x | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| P(x) | $1/8$ | $3/8$ | $3/8$ | $1/8$ |

**2.** Find the probability distribution for the product of dots appeared when two dice are thrown once.
$(Ans: see table)$

| x | 1 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| P(x) | $1/16$ | $2/16$ | $3/16$ | $4/16$ | $3/16$ | $2/16$ | $1/16$ |

**3.** A number is picked at random from the set of integers $\{1,2,3,...,100\}$. Suppose X represents the remainder after dividing the number by 7. Find the probability distribution of X.
$(Ans: see table)$

| x | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| P(x) | 0.14 | 0.15 | 0.15 | 0.14 | 0.14 | 0.14 | 0.14 |

**4.** Determine the constant *k* in each case of the following cases if the probability function of a random variable X is given by

(a) $P(X = x) = k(x-2),\ x = 3,4,5,6$

(b) $P(X = x) = k|x-2|,\ x = -1,0,1,3$

(c) $P(X = x) = k\ {}^{2}C_{x}\ {}^{3}C_{3-x},\ x = 0,1,2$

$(Ans: 1/10, 1/7, 1/10)$

**5.** The discrete random variable X has the given probability distribution

| x | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| P(x) | 0.2 | 0.25 | 0.4 | a | 0.05 |

Then find

(a) the value of a.

(b) $P(1 \le X \le 3)$

(c) $P(X > 2)$

(d) $P(2 < X < 5)$

$(Ans: 0.1, 0.85, 0.55, 0.5)$

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***DISTRIBUTION FUNCTION FOR DISCRETE RANDOM VARIABLE***

**6.** The probability distribution of X, the number of imperfections per 10 meters of a synthetic fabric in rolls of uniform width, is given by

| X | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| P(x) | 0.41 | 0.37 | 0.16 | 0.05 | 0.01 |

Construct the cumulative distribution function of X.

$(Ans : see table)$

| X | $x<0$ | $0\le x<1$ | $1\le x<2$ | $2\le x<3$ | $3\le x<4$ | $x\ge 4$ |
|---|---|---|---|---|---|---|
| F(x) | 0 | 0.41 | 0.78 | 0.94 | 0.99 | 1 |

**7.** A random variable X is defined by the function

$$P(X=x)={}^4C_x(0.6)^x(0.4)^{4-x},\ x=0,1,2,3,4$$

(a) Construct the probability distribution for X
(b) Construct the distribution function for X.
$(Ans : see table)$

| X | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| P(x) | 0.0256 | 0.1536 | 0.3456 | 0.3456 | 0.1296 |

| X | $x<0$ | $0\le x<1$ | $1\le x<2$ | $2\le x<3$ | $3\le x<4$ | $x\ge 4$ |
|---|---|---|---|---|---|---|
| F(x) | 0 | 0.0256 | 0.1792 | 0.5248 | 0.8704 | 1 |

**8.** The following table shows the distribution function $F(x)$ of the random variable X as shown in the table

| X | $x\le 1$ | $x\le 2$ | $x\le 3$ | $x\le 4$ |
|---|---|---|---|---|
| F(x) | $1/8$ | $3/8$ | $3/4$ | 1 |

Find the following
(a) Probability distribution of the random variable X
(b) $P(1\le X\le 3)$
(c) $P(X\le 2)$
(d) $P(X<3)$
$(Ans : see table)$

| X | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| P(x) | $1/8$ | $1/4$ | $3/8$ | $1/4$ |

$(Ans : 3/4, 3/8, 3/8)$

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**9.** For a discrete random variable X the cumulative distribution function $F(x)$ is shown in the table

| X | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| F(x) | 0.13 | 0.54 | 0.75 | 1 |

Find the following using probability distribution and distribution function tables

(a) $P(X = 2)$

(b) $P(X > 1)$

(c) $P(X \geq 3)$

(d) $P(X \geq 3)$

(e) $P(X < 2)$

$(Ans : 0.41, 0.87, 0.46, 0.75, 0.13)$

**10.** The cumulative probabilities for X are given in the following table, where X takes the values 0,1,2,….12.

| X | F(x) |
|---|---|
| 0 | 0.0115 |
| 1 | 0.0692 |
| 2 | 0.2061 |
| 3 | 0.4114 |
| 4 | 0.6296 |
| 5 | 0.8042 |
| 6 | 0.9133 |
| 7 | 0.9679 |
| 8 | 0.9900 |
| 9 | 0.9974 |
| 10 | 0.9994 |
| 11 | 0.9999 |
| 12 | 1 |

Use table to find the following

(a) $P(X \leq 8)$

(b) $P(X = 5)$

(c) $P(X \geq 4)$

(d) $P(3 < X \leq 7)$

(e) $P(1 \leq X < 9)$

$(Ans : 0.9900, 0.1746, 0.5886, 0.5565, 0.9785)$

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## ***EXPECTATION AND VARIANCE OF DISCTRETE RANDOM VARIABLES***

**11.** (a) The probability distribution for the random variable X is shown in the table.

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(X=x)$ | $1/6$ | $1/12$ | $1/4$ | $1/3$ | $1/6$ |

Find $E(X)$.

(b) The discrete random variable X has p.d.f $P(X=x)=kx \ \ for \ x=1,2,3,4,5$

Where $k$ is constant. Find $E(X)$.

$(Ans: 2.25, 11/3)$

**12.** A bag contains 5 black counters and six red counters. Two counters are drawn , one at a time, and not replaced. Let X be the number of red counters drawn.

Find $E(X)$.

$(Ans: 12/11)$

**13.** A discrete random variable X can take values 10 and 20 only. If $E(X)=16.$ write out the probability distribution of X.

$(Ans: see table)$

| $x$ | 10 | 20 |
|---|---|---|
| $P(X=x)$ | 0.4 | 0.6 |

**14.** A disk drive manufacturer sells storage devices with capacities of one terabyte, 500 gigabytes, and 100 gigabytes with probabilities 0.5, 0.3, and 0.2, respectively. The revenues associated with the sales in that year are estimated to be \$50 million, \$25 million, and \$10 million, respectively. Let *X* denote the revenue of storage devices during that year. Using probability mass function, find

(a) $E(X)$ (b) $Var(X)$

$(Ans: E(X)=34.5, V(X)=267.25)$

**15.** The discrete random variable X can take values $0,1,2 \ and \ 3 \ only$ . Given that $P(X \le 2)=0.9$, $P(X \le 1)=0.5$ and $E(X)=1.4$, then find

(a) $V(X)$

(b) $S.D(X)$

$(Ans: 0.84, 0.92)$

**16.** The table below shows the probability distribution for a random variable X.

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $P(X=x)$ | $c$ | $c^2$ | $c^2+c$ | $3c^2+2c$ |

Calculate

(a) $c$ (b) $E(X)$ (c) $Var(X)$

$(Ans: 0.2, 2.08, 1.35)$

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**17.** The discrete random variable X has p.d.f $P(X=x)$ for $x=1,2,3$

| $x$ | 1 | 2 | 3 |
|---|---|---|---|
| $P(X=x)$ | 0.2 | 0.3 | 0.5 |

Find

(a) $E(X)$

(b) $E(X^2)$

(c) $Var(X)$

$(Ans: 2.3, 5.9, 0.61)$

**18.** The discrete random variable X has p.d.f $P(X=x)=k$ for $x=1,2,3,4,5,6$. Find

(a) $E(X)$

(b) $E(X^2)$

(c) $E(3X+4)$

(d) $Var(X)$

(e) $Var(3X+4)$

$(Ans: 3.5, 15.67, 14.5, 2.92)$

**19.** The discrete random variable X has probability function given by

$$p(x)=\begin{cases}\left(\frac{1}{2}\right)^x & x=1,2,3,4,5\\ c & x=6\\ 0 & otherwise\end{cases}$$

Where $c$ is a constant, than find

(a) the value of $c$

(b) mean of X

(c) Variance of X.

$(Ans: 1/32, 1.96875,\quad)$

**20.** The discrete random variable X has p.d.f $P(X=x)=k|x|$ where x takes the values $x=-3,-2,-1,0,1,2,3$, Than find

(a) the value of $k$

(b) $E(X)$

(c) $E(X^2)$

(d) Standard deviation of X

(e) $E(2X-3)+Var(2X-3)$

$(Ans: 1/12, 0, 6, 2.45,\quad)$

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